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SOCKET ASSEMBLY

Abstract

The present disclosure provides a socket assembly thereof. The socket assembly includes a body, a plurality of shield components and a plurality of retractable probes. The body has a plurality of through holes and a plurality of recesses. The through holes accommodate the retractable probes. At least one recess is formed between any adjacent two through holes. The shield components are inserted in the recesses.

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Background/Summary

BACKGROUND OF THE PRESENT DISCLOSURE

Field of the Present Disclosure

[0001] The present disclosure relates to socket assemblies, and more particularly to a socket assembly having a shielding function and adapted for use with a chip test socket.

Description of the Prior Art

[0002] Upon completion of the packaging of chips, the packaged chips have to undergo a test to screen out unsatisfactory packaged chips. In general, the packaged chips undergo a test with a socket equipped with test probes, the test probes are usually elastic probes commonly known as pogo pins. One end of each pogo pin is in contact with a circuit board on the socket, and the other end of the pogo pin is in contact with the pins of a packaged chip, so as to perform the test.

[0003] The socket has through holes for receiving the pogo pins. However, the distance between the through holes is so short that signals associated with adjacent pogo pins during the test interfere with each other to cause the detriment of the precision of the test results. The interference is especially severe during a high-frequency signal test. Therefore, it is imperative to reduce signal interference and thereby enhance testing stability and precision.

[0004] The above description of the “prior art” merely serves to reveal background technology but is not intended to acknowledge that the above description of the “prior art” discloses the subject matters of the disclosure, constitutes the prior art of the disclosure, or is part of the disclosure.

SUMMARY OF THE PRESENT DISCLOSURE

[0005] In view of the aforesaid conventional issue of test signal interference, present disclosure provides a socket assembly.

[0006] An embodiment of the present disclosure provides a socket assembly used with a chip test socket, the socket assembly comprising a body, a plurality of first shield components, a plurality of second shield components and a plurality of pogo pins. The body has an upper surface, a lower surface opposite to the upper surface, a plurality of first through hole arrays, a plurality of second through hole arrays, a plurality of first grooves and a plurality of second grooves. Each first through hole array comprises a plurality of first through holes aligned in a first direction. Each second through hole array comprises a plurality of second through holes aligned in a second direction. Each first groove extends in the first direction and is disposed between any two adjacent first through hole arrays. Each second groove extends in the second direction and is disposed between any two adjacent second through hole arrays. Each first shield component has a first surface, and the plurality of first shield components are insertedly disposed in the plurality of first grooves. Each second shield component has a second surface, and the plurality of second shield components are insertedly disposed in the plurality of second grooves. The plurality of pogo pins are disposed in the plurality of first through holes and the plurality of second through holes. A first distance is defined between the first surface of each first shield component and the lower surface. A second distance is defined between the second surface of each second shield component and the lower surface.

[0007] Another embodiment of the present disclosure provides a socket assembly comprising a body, a plurality of shield components and a plurality of pogo pins. At least a first portion of a plurality of through holes form a plurality of first through hole arrays. At least a second portion of the plurality of through holes form a plurality of second through hole arrays. At least a first portion of a plurality of grooves are disposed between any two adjacent first through hole arrays. At least a second portion of the plurality of grooves are disposed between any two adjacent second through hole arrays. The plurality of shield components are insertedly disposed in the plurality of grooves. The plurality of pogo pins are disposed in the plurality of through holes.

[0008] Therefore, a socket assembly of the present disclosure includes shield components for shielding signal interference between pogo pins during a test to effectively shield signal interference and thereby enhance testing stability and precision.

[0009] The technical features and advantages of the disclosure are described generally and extensively above to render the detailed description of the disclosure below comprehensible. Other technical features and advantages of the subject matters of the claims of the disclosure are described below. Persons skilled in the art understand that concepts and specific embodiments presented below can be easily used to amend or design any other structures or manufacturing processes and thereby achieve the same objective as the disclosure. Persons skilled in the art also understand that the aforesaid equivalent constructions cannot depart from the spirit and scope defined in the appended claims of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Persons skilled in the art can gain insight into the disclosure by referring to the embodiments, claims and drawings of the disclosure. In the drawings, like components are denoted by like reference numerals.

[0011] FIG. 1A is a perspective view of a socket assembly according to some embodiments of the present disclosure.

[0012] FIG. 1B is a perspective view of the socket assembly according to some embodiments of the present disclosure.

[0013] FIG. 1C is a cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0014] FIG. 1D is a cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0015] FIG. 1E is a perspective view of a body according to some embodiments of the present disclosure.

[0016] FIG. 1F is a perspective view of the body according to some embodiments of the present disclosure.

[0017] FIG. 1G is a perspective view of the body according to some embodiments of the present disclosure.

[0018] FIG. 1H is a cross-sectional view of the body according to some embodiments of the present disclosure.

[0019] FIG. 1I is a cross-sectional view of the body according to some embodiments of the present disclosure.

[0020] FIG. 2A is a perspective view of a socket assembly according to some embodiments of the present disclosure.

[0021] FIG. 2B is a perspective view of the socket assembly according to some embodiments of the present disclosure.

[0022] FIG. 2C is a cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0023] FIG. 2D is a cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0024] FIG. 2E is a cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0025] FIG. 2F is a perspective view of the body according to some embodiments of the present disclosure.

[0026] FIG. 2G is a perspective view of the body according to some embodiments of the present disclosure.

[0027] FIG. 2H is a perspective view of the body according to some embodiments of the present disclosure.

[0028] FIG. 2I is a cross-sectional view of the body according to some embodiments of the present disclosure.

[0029] FIG. 2J is a cross-sectional view of the body according to some embodiments of the present disclosure.

[0030] FIG. 2K is a cross-sectional view of the body according to some embodiments of the present disclosure.

[0031] FIG. 2L is a cross-sectional view of shield components according to some embodiments of the present disclosure.

[0032] FIG. 2M is a perspective view of a socket assembly according to some embodiments of the present disclosure.

[0033] FIG. 2N is a cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0034] FIG. 2O is a cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0035] FIG. 2P is an enlarged, cross-sectional view of the socket assembly according to some embodiments of the present disclosure.

[0036] FIG. 3 is a perspective view of a socket assembly according to some embodiments of the present disclosure.

[0037] FIG. 4A is a cross-sectional view of a socket assembly according to some embodiments of the present disclosure.

[0038] FIG. 4B is a cross-sectional view of a socket assembly according to some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0039] The description below is accompanied by drawings that are incorporated into and constitute a part of the specification to illustrate the embodiments of the disclosure. However, the disclosure is not limited to the embodiments. The embodiments described below may be appropriately integrated to attain any other embodiments.

[0040] The purpose of the expressions “an embodiment,” “embodiment,” “exemplary embodiment,” “other embodiments” and “another embodiment” used herein is to state that the embodiments of the disclosure can include specific features, structures or characteristics. However, not every embodiment must include the specific features, structures or characteristics. Furthermore, repeated use of the expression “in an embodiment” does not necessarily imply referring to the same embodiment, as the possibility of referring to the same embodiment should not be ruled out. Socket assemblies mentioned herein are mainly for use in chip tests performed in the course of semiconductor testing.

[0041] The description below goes into detail about steps and structures in order for the disclosure to be fully comprehensible. Obviously, the embodiments of the disclosure do not place any limitations on specific details well known among persons skilled in the art. In addition, well-known structures and steps are not reiterated herein such that no limitations are unnecessarily placed on the disclosure. Preferred embodiments of the disclosure are described in detail below. However, in addition to the detailed description, the disclosure can be extensively implemented in other embodiments. The scope of the disclosure is defined by the appended claims rather than restricted to the detailed description.

[0042] The description below provides different embodiments or examples for use in implementing different features of the disclosure. Specific embodiments or examples of components and arrangements are described in detail herein to simplify the contents of the disclosure. The specific embodiments and examples are illustrative rather than restrictive of the disclosure. For instance, dimensions of components are not restricted to disclosed ranges or values but are based on process criteria and/or expected nature of devices. In the description below, an embodiment of forming a first feature on or above a second feature includes any embodiments in which the first and second

features thus formed are in direct contact with each other or includes any embodiments in which an additional feature is formed between the first feature and the second feature, allowing the first and second features to be not in direct contact with each other. For the sake of conciseness and clarity, the accompanying drawings are not drawn to scale. For the sake of conciseness, some layers/features are omitted from the accompanying drawings.

[0043] In addition, for explanatory purposes, expressions about spatial relationships, such as “beneath,” “below,” “lower,” “above” and “upper,” are used herein to explain the relationship between one component (or feature) and another component (or feature), as shown in the accompanying drawings. The expressions about spatial relationships are intended to indicate any other directions in which the components are used or operated in addition to the directions depicted in the accompanying drawings. The components may have any other directions (be rotated by 90 degrees or be aligned with another direction), and the expressions about spatial relationships can be construed accordingly.

[0044] An embodiment of the present disclosure relates to a socket assembly for a chip test socket, where the socket assembly includes a body, a plurality of shield components and a plurality of pogo pins. The plurality of pogo pins are configured to test high-frequency signals. The body is made from an insulating material, and has a plurality of through holes and a plurality of grooves. At least one of the grooves is disposed between two adjacent through holes. The plurality of through holes receive the plurality of pogo pins. The plurality of shield components are made from a conductive metallic material and are insertedly disposed in the plurality of grooves. Owing to the aforesaid technical features, the insulating property of the body precludes the electrical contact between the pogo pins, and metallic shield components are insertedly disposed between the through holes where the pogo pins are received in, so as to effectively shield signal interference associated with the pogo pins during testing, especially signal interference during the high-frequency signal testing. Therefore, when the socket assembly is mounted on a corresponding socket, the socket assembly can effectively shield signal interference to enhance testing stability and precision. The details of required structural features are illustrated by embodiments and described below.

[0045] Reference is made to FIG. 1A through FIG. 1D. FIG. 1A is a perspective view of a socket assembly **11** according to some embodiments of the present disclosure. FIG. 1B is another perspective view of the socket assembly **11** according to some embodiments of the present disclosure. FIG. 1C is a cross-sectional view of the socket assembly **11** according to some embodiments of the present disclosure. FIG. 1D is another cross-sectional view of the socket assembly **11** according to some embodiments of the present disclosure. Specifically, the socket assembly **11** includes a body **111**, a plurality of shield components **113** and a plurality of pogo pins **13**. The body **111** has an upper surface **111S1** and a lower surface **111S2** opposite to the upper surface **111S1**.

[0046] Reference is made to FIG. 1E through FIG. 1I. FIGS. 1E and 1F are perspective views of the body **111** according to some embodiments of the present disclosure. FIG. 1G is another perspective view of the body **111** according to some embodiments of the present disclosure. FIG. 1H is a cross-sectional view of the body **111** according to some embodiments of the present disclosure. FIG. 1I is another cross-sectional view of the body **111** according to some embodiments of the present disclosure. Specifically, the body **111** has multiple through holes **1110** and multiple grooves **1112**. At least a first portion of the through holes **1110** form a plurality of first through hole arrays **1110AR1**. The through holes **1110** in each first through hole array **1110AR1** are defined as first through holes and aligned with a first direction **D13**. At least a second portion of the through holes **1110** form a plurality of second through hole arrays **1110AR2**. The through holes **1110** in each second through hole array **1110AR2** are defined as second through holes and aligned with a second direction **D14**. In some embodiments, one of the first through holes and one of the second through holes can be the same through hole, whereas the first through holes are distinguished from the second through holes according to the angles of different through hole arrays.

[0047] In some embodiments, the multiple grooves **1112** include multiple first grooves **11120** and multiple second grooves **11122**. Each first groove **11120** is extended in the first direction **D13** and disposed between two adjacent first through hole arrays **1110AR1**. Each second groove **11122** is extended in the second direction **D14** and disposed between two adjacent second through hole arrays **1110AR2**.

[0048] In some embodiments, multiple shield components **113** include multiple first shield components **1131** and multiple second shield components **1133**. The first shield components **1131** are inserted into the first grooves **11120** from the upper surface **111S1**, respectively. The second shield components **1133** are inserted into the second grooves **11122** from the upper surface **111S1**, respectively. The through holes **1110** receive the pogo pins **13**. In some embodiments, the first shield components **1131** and second shield components **1133** are staggered with an engagement structure disposed between every two adjacent shield components.

[0049] In some embodiments, each first shield component **1131** has a first surface **1131S**, and each second shield component **1133** has a second surface **1133S**. The first surfaces **1131S**, the second surfaces **1133S** and the lower surface **111S2** substantially face the same direction. Upon the insertion of the first shield components **1131** and second shield components **1133** into the first and second grooves respectively, a first distance **D11** is defined between the lower surface **111S2** and each first surface **1131S**, and a second distance **D12** is defined between the lower surface **111S2** and each second surface **1133S**.

[0050] In some embodiments, the upper surface **111S1** and the lower surface **111S2** are substantially parallel, allowing the first surfaces **1131S** of the first shield components **1131** to be substantially coplanar, the second surfaces **1133S** of the second shield components **1133** to be substantially coplanar, the first surfaces **1131S** and the second surfaces **1133S** to be substantially coplanar, and the first distance **D11** to be substantially equal to the second distance **D12**, but the aforesaid technical features are not restrictive of the embodiments of present disclosure.

[0051] Therefore, owing to the structure of the socket assembly **11**, when the pogo pins **13** received in two adjacent through holes **1110** are used in the signal transmission during a chip test, the signal interference is effectively shielded by using the shield components **113** which are inserted into the grooves **1112** disposed between two adjacent through holes **1110**. Referring to FIG. 1A, for example, when the pogo pins **13A**, **13B** received in the through holes **1110A**, **1110B** are used in the signal transmission during a chip test, the signal interference is effectively shielded by using the shield components **113A** which are inserted into the groove **1112A** disposed between the through hole **1110A** and the through hole **1110B**.

[0052] Reference is made to FIG. 2A through FIG. 2E. FIG. 2A is a perspective view of a socket assembly **21** according to some embodiments of the present disclosure. FIG. 2B is another perspective view of the socket assembly **21** according to some embodiments of the present disclosure. FIG. 2C through FIG. 2E are cross-sectional views of the socket assembly **21** according to some embodiments of the present disclosure. Specifically, the socket assembly **21** includes a body **211**, a plurality of shield components **213** and a plurality of pogo pins **23**. The body **211** has an upper surface **211S1** and a lower surface **211S2** opposite to the upper surface **211S1**.

[0053] Reference is made to FIG. 2F through FIG. 2K. FIGS. 2F and 2G are perspective views of the body **211** according to some embodiments of the present disclosure. FIG. 2H is another perspective view of the body **211** according to some embodiments of the present disclosure. FIG. 2I through FIG. 2K are cross-sectional views of the body **211** according to some embodiments of the present disclosure. Specifically, the body **211** is block-shaped (for example, in a shape of a cubic block) and has multiple first through hole arrays **2110AR1** and multiple second through hole arrays **2110AR2**. Each first through hole array **2110AR1** includes a plurality of first through holes **2110-1** aligned with a first direction **D23**. Each second through hole array **2110AR2** includes a plurality of second through holes **2110-2** aligned with a second direction **D24**. In these embodiments, the first direction **D23** and the second direction **D24** are basically perpendicular to each other, however, the

present disclosure is not limited thereto. In some embodiments, one of the first through holes **2110-1** and one of the second through holes **2110-2** can be the same through hole, whereas the first through holes are distinguished from the second through holes according to the angles of different through hole arrays.

[0054] The body **211** has a plurality of first grooves **21120** and a plurality of second grooves **21122**. Each of the first grooves **21120** is extended in the first direction **D23** and disposed between two adjacent first through hole arrays **2110AR1**. Each of the second grooves **21122** is extended in the second direction **D24** and disposed between two adjacent second through hole arrays **2110AR2**. In some embodiments, the first grooves **21120** and the second grooves **21122** are staggered.

[0055] In some embodiments, the plurality of shield components **213** include a plurality of first shield components **2131** and a plurality of second shield components **2133**. The plurality of first shield components **2131** are inserted into the plurality of first grooves **21120** of the body **211** from the upper surface **211S1**, respectively. The plurality of second shield components **2133** are inserted into the plurality of second grooves **21122** of the body **211** from the upper surface **211S1**, respectively. The first shield components **2131** and the second shield components **2133** are staggered. The plurality of first through holes **2110-1** and the plurality of second through holes **2110-2** receive the plurality of pogo pins **23**.

[0056] In some embodiments, each of the first shield components **2131** has a first surface **2131S**, and each of the second shield components **2133** has a second surface **2133S**. The first surfaces **2131S**, the second surfaces **2133S** and the lower surface **211S2** substantially face the same direction. The first grooves **21120** and the second grooves **21122** share a bottom surface **211S3** inside the body **211**. The body **211** has a thickness **D25** between the bottom surface **211S3** and the lower surface **211S2**. Upon the insertion of the first shield components **2131** into the corresponding grooves, the first surfaces **2131S** adjoin the bottom surface **211S3** of the body **211**, whereas a first distance **D21** is defined between each first surface **2131S** and the lower surface **211S2**. Upon the insertion of the second shield components **2133** into the corresponding grooves, the second surfaces **2133S** adjoin the bottom surface **211S3** of the body **211**, whereas a second distance **D22** is defined between each second surface **2133S** and the lower surface **211S2**. In some embodiments, the thickness **D25**, the first distance **D21** and the second distance **D22** are equal to one another.

[0057] In some embodiments, two first grooves **21120** and two second grooves **21122** are arranged in a pattern that looks like a pound sign (#), and a through hole (for example, the first through holes **2110-1** or the second through holes **2110-2**) is formed in each of nine cells of the pound sign. Given the pattern, at least one groove (for example, first groove **21120** or second groove **21122**) is disposed between any two adjacent through holes (for example, the first through holes **2110-1** and the second through holes **2110-2**). The first shield components **2131** inserted into the first grooves **21120** or the second shield components **2133** inserted into the second grooves **21122** are disposed between the pogo pins **23** in any two adjacent through holes (for example, the first through holes **2110-1** and the second through holes **2110-2**).

[0058] In some embodiments, each of the first grooves **21120** has a first opening **21120X** on the upper surface **211S1** of the body **211** and has a plurality of second openings **21120Y** on the lower surface **211S2** of the body **211**. The second openings **21120Y** are formed along the direction of the extension of the first grooves **21120** and spaced apart from each other. The dimension of each of the second openings **21120Y** is smaller than the dimension of the first opening **21120X**. The first shield components **2131** have a plurality of protrusion portions **2131P**. When the first shield components **2131** are inserted into the first grooves **21120** via the first openings **21120X**, the protrusion portions **2131P** penetrate through the corresponding second openings **21120Y** of the first grooves **21120** and protrude from the lower surface **211S2** (i.e., the protrusion portions **2131P** protrude from and exceed the lower surface **211S2**).

[0059] In some embodiments, each of the second grooves **21122** has a third opening **21122X** on the upper surface **211S1** of the body **211** and each have a plurality of fourth openings **21122Y** on the

lower surface **211S2** of the body **211**. The fourth openings **21122Y** are formed along the direction of the extension of the second grooves **21122** and spaced apart from each other. The dimension of each of the fourth openings **21122Y** is smaller than the dimension of the third opening **21122X**. The second shield components **2133** have a plurality of protrusion portions **2133P**. When the second shield components **2133** are inserted into the second grooves **21122** via the third openings **21122X**, the protrusion portions **2133P** penetrate through the corresponding fourth openings **21122Y** of the second grooves **21122** and protrude from the lower surface **211S2** (i.e., the protrusion portions **2133P** protrude from and exceed the lower surface **211S2**).

[0060] In some embodiments, each first protrusion portion **2131P** of the first shield components **2131** has a third surface **2131PS**, and all the third surfaces **2131PS** are substantially coplanar. Each second protrusion portion **2133P** of the second shield components **2133** has a fourth surface **2133PS**, and all fourth surfaces **2133PS** are substantially coplanar. In some embodiments, all of the third surfaces **2131PS** and all of fourth surfaces **2133PS** are substantially coplanar.

[0061] Referring to FIG. 2L, which is shown a cross-sectional view of the first shield components **2131** and the second shield components **2133** according to some embodiments of the present disclosure. Specifically, each first shield component **2131** has a plurality of first joints **21310**, and each second shield component **2133** has a plurality of second joints **21330**. When the first shield components **2131** and the second shield components **2133** are staggered, the first joints **21310** receive the second shield components **2133**, and the second joints **21330** receive the first shield components **2131**.

[0062] In some embodiments, the first joints **21310** and the second joints **21330** are slots and engaged with each other. In these embodiments, the depth of the first joints **21310** is substantially equal to half the height of the first shield components **2131**, and the depth of the second joints **21330** is substantially equal to half the height of the second shield components **2133**. When the first shield components **2131** and the second shield components **2133** have the same height, the height of the first and second shield components **2131**, **2133** engaged with each other is equal to the original height of the first shield components **2131** or the second shield components **2133**.

[0063] It should be noted that these embodiments are not restrictive of the implementation of the first shield components **2131** and the second shield components **2133**. In some embodiments, the first shield components **2131** and the second shield components **2133** have the same height, and the sum of the depth of the first joint **21310** and the depth of the second joint **21330** is substantially equal to the height of the first shield components **2131** or the second shield components **2133**, whereas the height of the first and second shield components **2131**, **2133** engaged with each other is equal to the original height of the first shield components **2131** or the second shield components **2133**. In some embodiments, the first shield components **2131** and the second shield components **2133** are monolithic (one-piece formed).

[0064] Therefore, owing to the structure of the socket assembly **21**, when the pogo pins **13** received in two adjacent through holes (for example, the first through hole **2110-1** or the second through hole **2110-2**) are used in the signal transmission during a chip test, the signal interference is effectively shielded by the first shield components **2131** or the second shield components **2133** disposed between two adjacent through holes (for example, the first through hole **2110-1** and the second through hole **2110-2**), in which the first shield components **2131** and the second shield components **2133** are inserted into the first grooves **21120** and the second grooves **21122**, respectively. The protrusion portions **2131P**, **2133P** of the first shield components **2131** and the second shield components **2133** increase the shielding range between the pogo pins **23** and enhance signal interference shielding.

[0065] Referring to FIG. 2A, for example, when the pogo pin **23A**, **23B** received in through holes **2110A**, **2110B** are used in the signal transmission during a chip test, the signal interference is effectively shielded by the second shield components **2133A** inserted into second grooves **21122A** disposed between the through holes **2110A** **2110B**. Furthermore, the protrusion portions **2133P** of

the second shield components **2133A** increase the shielding range.

[0066] Reference is made to FIG. 2M through FIG. 2O. FIG. 2M is a perspective view of a socket assembly **21'** according to some embodiments of the present disclosure. FIG. 2N is a cross-sectional view of the socket assembly **21'** according to some embodiments of the present disclosure. FIG. 2O is another cross-sectional view of the socket assembly **21'** according to some embodiments of the present disclosure. Specifically, compared to the socket assembly **21**, the socket assembly **21'** further includes a circuit substrate **25**. The circuit substrate **25** is disposed on the upper surface **211S1** of the body **211** and electrically connected to the pogo pins **23** to transmit test-related signals. The other end of each pogo pin **23** is configured to contact an object under test (not shown). In some embodiments, the first shield components **2131**, the second shield components **2133** and the circuit substrate **25** are electrically connected so as to form a ground loop.

[0067] In some embodiments, the socket assembly **21'** includes a first shield component **2131'**. Compared with the first shield components **2131**, the first shield component **2131'** has a first fixing structure **2131F**. The circuit substrate **25** has a second fixing structure **251** corresponding to the shape of the first fixing structure **2131F**. The first fixing structure **2131F** and the second fixing structure **251** can be engaged with each other, allowing the first shield component **2131'** and the circuit substrate **25** to be firmly coupled together. It should be noted that, in these embodiments, the first fixing structure **2131F** is hook-shaped, and the second fixing structure **251** is notch-shaped corresponding to the hook-shaped first fixing structure **2131F**, however, the present disclosure is not limited thereto. Thus, in a variant embodiment, the second fixing structure **251** is hook-shaped, and the first fixing structure **2131F** is notch-shaped corresponding to the hook-shaped second fixing structure **251**. In some embodiments, the first fixing structure **2131F** of the first shield component **2131'** and the second fixing structure **251** of the circuit substrate **25** are electrically connected so as to form a ground loop. Similarly, the second shield component **2133** has the aforesaid fixing structure and thus can be engaged with the circuit substrate **25**, and related details are not reiterated for the sake of brevity.

[0068] Referring to FIG. 2P, which is shown an enlarged view of a portion of FIG. 2N. In some embodiments, a third distance **D26** is defined between an end **2133E** of each protrusion portion **2133P** of the second shield components **2133** and the lower surface **211S2** (i.e., between the third surface **2131PS** and the lower surface **211S2**), and a fourth distance **D27** is defined between the lower surface **211S2** and an adjacent one of the pogo pins **23** received in the through holes respectively. The fourth distance **D27** is greater than the third distance **D26**, and the difference therebetween is greater than or equal to the compression stroke of the pogo pins **23** so as to prevent the compressed pogo pins **23** from engaging with the protrusion portions **2133P** of the second shield components **2133**. Similarly, the first shield components **2131** also has the aforesaid technical features to prevent the compressed pogo pins **23** from engaging with the protrusion portions **2131P** of the first shield components **2131**, and related details are not reiterated for the sake of brevity.

[0069] Referring to FIG. 3, which is shown a perspective view of a socket assembly **31** according to some embodiments of the present disclosure. Specifically, the socket assembly **31** includes a body **311** and multiple shield components **313**. The body **311** has an upper surface **311S1**. The body **311** has multiple through holes **3110** and multiple independent grooves **3112**. The grooves **3112** are not in communication with each other. Each groove **3112** is disposed between two adjacent through holes **3110**. The shield components **313** are insertedly disposed in the grooves **3112**. The through holes **3110** receive multiple pogo pins **33**.

[0070] Referring to FIG. 4A, which is shown a perspective view of a socket assembly **41** according to some embodiments of the present disclosure. Specifically, the socket assembly **41** includes a body **411** and a plurality of shield components **413**. The body **411** has an upper surface **411S1**. The body **411** has a plurality of through holes **4110** and a plurality of grooves **4112**. Two grooves **4112**

are disposed between any two adjacent through holes **4110**. The plurality of shield components **413** are insertedly disposed in the plurality of grooves **4112**. The plurality of through holes **4110** receive a plurality of pogo pins **43**.

[0071] Referring to FIG. **4B**, there is shown a perspective view of a socket assembly **41'** according to some embodiments of the present disclosure. Specifically, the difference between the socket assembly **41'** and the socket assembly **41** is that the grooves **4112** of the socket assembly **41'** and the shield components **413** are interleaved, which is also able to achieve the effect of shielding.

[0072] In some embodiments, the width of groove is smaller than the thickness of the corresponding pogo pin. In some embodiments, the material of the body includes an insulating material for precluding electrical contact between the pogo pins, and the body is monolithic (for example, by injection molding) to form a block-shaped solid that has through holes and grooves. In some embodiments, the material which the shield components are made from includes a metallic material (for example, an electrically conductive metallic material) for shielding signal interference.

[0073] In these embodiments, the shield components are disposed between any two pogo pins, but the present disclosure is not limited thereto. Well-known among persons skilled in the art is that, in some embodiments, it is not necessary for a signal generated from every pogo pin to be shielded, rendering it feasible to provide the shield components only between the pogo pins required to be shielded (for example, pogo pins for detecting high-frequency signals).

[0074] Furthermore, it should be noted that the pogo pins in the present disclosure are provided for illustrative purposes. Various probe pins are within the contemplated scope of the present disclosure.

[0075] In the embodiments, for the sake of brevity and ease of comprehension, although like components may be shown in the accompanying drawings, only one of the like components is denoted in the accompanying drawings, without imposing limitations on the disclosure. Persons skilled in the art should be able to perceive and comprehend like components shown in the accompanying drawings.

[0076] Therefore, the embodiments of the disclosure provide socket assemblies each comprising shield components embedded therein and adapted to achieve shielding between pogo pins so as to reduce signal interference in the course of testing and thereby effectively enhance testing stability and precision.

[0077] Although the disclosure and advantages thereof are described above, persons skilled in the art understand that various changes, replacements and substitutions may be made to the disclosure without departing from the spirit and scope defined in the appended claims of the disclosure. For instance, the aforesaid processes may be implemented with different methods and replaced with any other processes or a combination thereof.

[0078] The scope of the disclosure is not restricted to specific embodiments of any processes, machines, manufacturing, matter compositions, means, methods and steps described herein. The disclosure described herein enables persons skilled in the art to implement the disclosure with any existing or potential processes, machines, manufacturing, matter compositions, means, methods or steps having the same function or capable of achieving substantially the same result as disclosed in the aforesaid embodiments. Therefore, these processes, machines, manufacturing, matter compositions, means, methods and steps fall within the scope of the appended claims of the disclosure.

Claims

1. A socket assembly used with a chip test socket, the socket assembly comprising: a body having: an upper surface; a lower surface opposite to the upper surface; a plurality of first through hole arrays each comprising a plurality of first through holes aligned with a first direction; a plurality of

second through hole arrays each comprising a plurality of second through holes aligned with a second direction; a plurality of first grooves each extending in the first direction, wherein any two adjacent first through hole arrays has one of the first grooves disposed therebetween; and a plurality of second grooves each extending in the second direction, wherein any two adjacent second through hole arrays has one of the second grooves disposed therebetween; a plurality of first shield components each having a first surface and being inserted into the first grooves from the upper surface; a plurality of second shield components each having a second surface and being inserted into the second grooves from the upper surface; and a plurality of pogo pins disposed in the first through holes and the second through holes, wherein a first distance is defined between each of the first surfaces of the first shield components and the lower surface, wherein a second distance is defined between each of the second surfaces of the second shield components and the lower surface.

2. The socket assembly of claim 1, wherein the first surfaces are coplanar, and the second surfaces are coplanar.

3. The socket assembly of claim 2, wherein the first surfaces and the second surfaces are coplanar.

4. The socket assembly of claim 1, wherein the first distance is equal to the second distance.

5. The socket assembly of claim 1, wherein the first direction is perpendicular to the second direction, the first grooves and the second grooves are staggered, and the first shield components and the second shield components are staggered.

6. The socket assembly of claim 5, wherein each of the first shield components has a plurality of first joints, and the first joints receive the second shield components.

7. The socket assembly of claim 6, wherein each of the second shield components has a plurality of second joints, and the second joints receive the first shield components.

8. The socket assembly of claim 1, wherein each of the first grooves has a first opening on the upper surface of the body and a plurality of second openings on the lower surface of the body, wherein a dimension of each of the second openings is smaller than a dimension of the first opening, and each of the second grooves has a third opening on the upper surface of the body and a plurality of fourth openings on the lower surface of the body, wherein a dimension of each of the fourth openings is smaller than a dimension of the third opening.

9. The socket assembly of claim 8, wherein at least one of the first shield components has a plurality of first protrusion portions, the first protrusion portions are penetratingly disposed in the second openings of the first grooves, respectively, and the first protrusion portions protrude from the lower surface.

10. The socket assembly of claim 9, wherein at least one of the second shield components has a plurality of second protrusion portions, the second protrusion portions are penetratingly disposed in the fourth openings of the second grooves, respectively, and the second protrusion portions protrude from the lower surface.

11. The socket assembly of claim 10, wherein a third distance is defined between an end of each of the first and second protrusion portions and the lower surface, when a first pogo pin of the pogo pins is received in a corresponding one of through holes, a fourth distance is defined between an end of the first pogo pin and the lower surface, and a difference between the fourth distance and the third distance is greater than or equal to a compression stroke of the first pogo pin.

12. The socket assembly of claim 10, wherein each of the first protrusion portions has a third surface, the third surfaces of the first protrusions are coplanar, and each of the second protrusion portions has a fourth surface, the fourth surfaces of the second protrusion portions are coplanar.

13. The socket assembly of claim 12, wherein the third surfaces and the fourth surfaces are coplanar.

14. The socket assembly of claim 1, further comprising a circuit substrate disposed on the upper surface of the body, wherein the first shield components and the second shield components are electrically connected to the circuit substrate so as to form a ground loop.

- 15.** The socket assembly of claim 1, further comprising a circuit substrate disposed on the upper surface of the body, wherein the first shield components and the second shield components have a plurality of first fixing structures engaging with a plurality of second fixing structures of the circuit substrate.
- 16.** The socket assembly of claim 15, wherein the first fixing structures are electrically connected to the second fixing structures.
- 17.** The socket assembly of claim 1, wherein a width of the first grooves and a width of the second grooves are less than a thickness of the pogo pins.
- 18.** The socket assembly of claim 1, wherein the body is made from an insulating material, and a material of the first shield components and the second shield components comprises a conductive metallic material.
- 19.** The socket assembly of claim 1, wherein the first grooves and the second grooves share a bottom surface in the body, and a thickness is defined between the bottom surface in the body and the lower surface of the body.
- 20.** A socket assembly, comprising: a body having: a plurality of through holes, wherein at least a first portion of the through holes form a plurality of first through hole arrays, and at least a second portion of the through holes form a plurality of second through hole arrays; and a plurality of grooves, wherein at least a first portion of the grooves are disposed between any two adjacent first through hole arrays, and at least a second portion of the grooves are disposed between any two adjacent second through hole arrays; a plurality of shield components insertedly disposed in the grooves; and a plurality of pogo pins disposed in the through holes.
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